



BHARATIYA VIDYA BHAVAN'S

SARDAR PATEL INSTITUTE OF TECHNOLOGY

MUNSHI NAGAR , ANDHERI(W), MUMBAI 400058

Electronics and Telecommunication Department,

Academic Year- 2025-2026

Mobile Wireless Communication - Experiment 4A

Team : Dhruv Santosh Warik (UID: 2023200130)

Rajas Mangirish Wagle (UID: 2023200129)

Kevin Vaz (UID: 2023200127)

Kaustubh Vekhande (UID: 2023200128)

Branch : EXTC

Batch : B4

Aim: To study GSM Modem and AT Commands.

System Requirements: Hardware : GSM modem, PC, SIM card. Software : Hyper Terminal.

Introduction:

The Nokia 30 uses GSM (Global System for Mobile Communications) technology to provide wireless connectivity. GSM allows the terminal to send and receive voice, data, and text messages over mobile networks.

By using GSM features such as GPRS (for internet access), SMS (for text communication), and USSD (for network services), the Nokia 30 can connect a computer or device to the mobile network. This makes it useful in applications like:

- Remote monitoring and control of devices.
- Sending alerts or notifications through SMS.
- Wireless internet access for PCs and equipment.
- Point-of-sale terminals and other business solutions that need mobile connectivity.

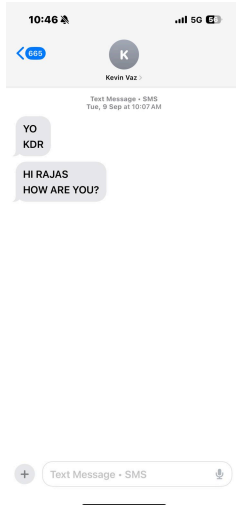
In short, GSM in the Nokia 30 acts as a bridge between devices and the mobile network, enabling reliable wireless communication anywhere within GSM coverage.

```
AT
OK
AT+CSQ
+CSQ: 24,0

OK

AT+CMGF=1
OK
AT+CMGS="+917506650490"
> HI RAJAS
> HOW ARE YOU?
+CMGS: 34

OK
-
```



```
TEMP - HyperTerminal
File Edit View Call Transfer Help
[Icons]

OK
AT+CGMM
SIMCOM_SIM800

OK
AT+CGMR
Revision:1308B09SIM800M32

OK
AT+CGSM
ERROR
AT+CGSM
ERROR

AT+CSCS
ERROR
AT+CSCS
ERROR
AT+CSCS="GSM"
OK
AT+CTMI
404205053757040

OK
AT+CCID
899120005053757040f

OK
AT+CMGF=1
OK
AT+CMGS="+917506650490"
> YO
> KDR
>
+CMGS: 32
OK
_
```

Connected 0:03:35 | Auto detect | Auto detect | SCROLL | CAPS | NUM | Capture | Print echo

OK
AT+CMGF=0
OK
ATD+917506650490;
OK

+CMT: "",137
07911989020475820C14D0566DCB9A1E06A5C5D614001052909001900522824176594E0F81B2EFBA
1C644D83AA6E76BA9DA697C9A07198CD4EBBCF207878BC0615B1D0A4B43805BDDDA058AC352DC35B
B29A0B447D83C66F373DEDAE974165B7FA9D4FBBCF207878BC0689CBEEB2394D9FB340F2F2181D96
9FCB20E7F30A42D3E9F0B9EEF512A7E92E76FE654DB7AE4429

HI
NO CARRIER

-

Connected 0:06:01 Auto detect Auto detect SCROLL CAPS NUM Capture Print echo

AT
OK
AT+CSQ
+CSQ: 24,0

OK
AT+CGMM
SIMCOM_SIM800

OK
AT+CGMR
Revision:1308B09SIM800M32

OK

-

Connected 0:01:46 Auto detect 9600 8-N-1 SCROLL CAPS NUM Capture Print echo

OK
AT+CMGF=0
OK
ATD+917506650490;
OK

+CMT: "",137
07911989020475820C14D0566DCB9A1E06A5C5D614001052909001900522824176594E0F81B2EFBA
1C644D83AA6E76BA9DA697C9A07198CD4EBBCF207878BC0615B1D0A4B43805BDDDA058AC352DC35B
B29A0B447D83C66F373DEDAE974165B7FA9D4FBBCF207878BC0689CBEEB2394D9FB340F2F2181D96
9FCB20E7F30A42D3E9F0B9EEF512A7E92E76FE654DB7AE4429

HI
NO CARRIER

+CMT: "",86
07911989020475820C14D0566DCB9A1E06A5C5D61400105290900101232248CCF09CDE1A86D96C1D
0CA683C17432180B344487E5E7B24E3A87B9603016A8194EBB5BC2305B279DC35C30180B546542C3
E375AB8887EB62B1D6B40C6FC96A

AT+CGSN
864495038220924

OK
AT+CSCS="GSM"

OK
AT+CIMI
404205053757040

OK
AT+CCID
8991200050537570405f

OK

OK
ATDL
OK
AT+CMUT=0
OK

NO CARRIER
AT+CMGF=1
OK
AT+CMGS="+919324835687"
> HI DHRUV SANTOSH WARIK
>
+CMGS: 35

OK
AT+CREG?
+CREG: 0,1

OK
AT+CNEG=1
ERROR
AT+CENG=1
OK

AT+CNEG=1
ERROR
AT+CENG=1
OK
AT+CENG?
+CENG: 1,0

+CENG: 0,"0706,53,00,404,20,55,fcd6,05,00,716f,255"
+CENG: 1,"0696,38,40,404,20,716f"
+CENG: 2,"0701,35,12,404,20,716f"
+CENG: 3,"0702,30,07,404,20,716f"
+CENG: 4,"0703,25,28,404,20,716f"
+CENG: 5,"0705,32,35,,,0000"
+CENG: 6,"0707,32,00,,,0000"

OK

```
AT+COPS
ERROR
AT+COPS?
+COPS: 0,0,"Orange"

OK
```

Connected 0:12:28 Auto detect 9600 8-N-1 SCROLL CAPS NUM Capture Print echo

Link to Videos:

https://drive.google.com/drive/folders/1QMXKUqE6ds_0thNjqjG5x6Y0_Eonyuew?usp=sharing

Conclusion:

We have successfully implemented the GSM AT commands using the GSM modem as hardware by inserting a Vodafone SIM and on the software side, we had the Hyperterminal, which played a crucial role in executing the AT commands. Along with that we easily implemented the messaging and call from the SIM using the AT commands. Overall, we learnt about the AT commands in detail along with implementing it on hardware.